AP CALCULUS AB - UNIT 1

Guided Notes 1.14: Connecting Infinite Limits and Vertical Asymptotes

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we interpret positive and negative infinite limits as directional unbounded behavior?

- Interpret positive and negative infinite limits as directional unbounded behavior.
- Determine one-sided signs near denominator zeros.
- Use factor multiplicity and cancellation to distinguish holes from vertical asymptotes.
- Connect formulas, sign charts, tables, graphs, and asymptote notation.

Key Knowledge, Vocabulary, and Notation

Vertical asymptote	$x = c$ is a vertical asymptote if at least one one-sided limit is $+\infty$ or $-\infty$.
Odd multiplicity	A sign-changing factor usually makes opposite branch signs.
Even multiplicity	A nonnegative small factor usually gives the same branch sign.
Cancellation first	A canceled denominator factor creates a hole, not a vertical asymptote.
Numerator sign	After simplification, combine numerator sign with every denominator factor sign on each side.

Concept Development and Strategy

1. Factor and cancel before building the sign chart.
2. For each candidate denominator zero, choose a test point just left and just right.
3. State both one-sided limits; merely listing the asymptote line is incomplete when signs are requested.

Worked Example: Odd pole

Analyze $f(x) = \frac{2}{x-3}$ near $x = 3$.

Answer and Reasoning

The numerator is positive. The denominator is negative on the left and positive on the right, so the limits are $-\infty$ and $+\infty$.

Worked Example: Even pole

Analyze $g(x) = \frac{-5}{(x+1)^2}$ near $x = -1$.

Answer and Reasoning

The denominator is positive on both sides and tends to 0, while the numerator is negative. Both one-sided limits are $-\infty$.

Worked Example: Hole before pole

Analyze $h(x) = \frac{(x-2)(x+3)}{(x-2)(x-4)}$.

Answer and Reasoning

Cancel $x - 2$. There is a removable hole at 2 and a vertical asymptote at 4. Near 4, $x + 3 > 0$, so the left limit is $-\infty$ and the right limit is $+\infty$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Find the vertical asymptote of $1/(x-6)$.

2. Find $\lim_{x\to 6^-} 1/(x-6)$.

3. Analyze $\frac{-3}{x+4}$ at $x = -4$.

4. Analyze $\frac{x+1}{(x-2)^2}$ at $x = 2$.

5. Find all vertical asymptotes and one-sided signs of $\frac{x+2}{(x-1)(x+3)}$.

6. For $f(x) = \frac{x-a}{(x-a)^m}$, classify the behavior at a for integer $m \geq 1$.

7. Construct a rational function with a hole at 1, an even pole at -2 opening downward, and an odd pole at 4 with left $+\infty$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. For $f(x) = 1/(x-2)^2$, the one-sided limits at 2 are

(A) both $-\infty$
(B) both $+\infty$
(C) left $-\infty$, right $+\infty$
(D) finite

2. Which factor behavior produces opposite branch signs, assuming other factors stay nonzero?

(A) An even-multiplicity denominator factor
(B) An odd-multiplicity denominator factor
(C) A canceled factor
(D) A positive constant only

3. A calculator graph of $\frac{x^2-4}{(x-2)^2}$ near 2 shows opposite unbounded branches. The simplified local form is

(A) $x+2$
(B) $1/(x-2)$
(C) $(x-2)^2$
(D) 0

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Prove that if a simplified rational function has a denominator zero of odd multiplicity and a nonzero numerator at c, then its one-sided infinite limits have opposite signs.